FORMAL CONSEQUENCES OF THE MULTIPLICATIVE SYSTEM AXIOMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1418-1458, following the statement at 1404-1416.

Editorial note. No omissions: both parts display their diagrams; RMS2 is marked dual by the author.

Allowed standard prerequisites (editorial).

- Zero triangles. lemma-third-object-zero (same chapter, cited at ~1421). The displayed rows are distinguished triangles.

Source reference. Original reference 04VC: focused-seeds/geometry/S9-der-localization-conditions/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 05R2: sources/stacks-derived/pinned-original-source.tex, source lines 1374–1374. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05QR: sources/stacks-derived/pinned-original-source.tex, source lines 833–833. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. Let $S \subset Arrows(\mathcal{D})$.*

(1) *If S contains all identities and MS6 holds (Definition 05R2), then every isomorphism of $\mathcal{D}$ is in S.*
(2) *If MS1, MS5 (Categories, Definition 04VC) and MS6 hold, then MS2 holds.*

Proof. Assume S contains all identities and MS6 holds. Let $f : X \to Y$ be an isomorphism of $\mathcal{D}$. Consider the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & X & \xrightarrow[1]{} & X & \longrightarrow & 0[1] \\ \downarrow{\scriptstyle 1} & & \downarrow{\scriptstyle 1} & & \vdots & & \downarrow{\scriptstyle 1[1]} \\ 0 & \longrightarrow & X & \xrightarrow{f} & Y & \longrightarrow & 0[1] \end{array}$$

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

The rows are distinguished triangles by Lemma 05QR. By MS6 we see that the dotted arrow exists and is in S, so f is in S.

Assume MS1, MS5, MS6. Suppose that $f : X \to Y$ is a morphism of $\mathcal{D}$ and $t : X \to X'$ an element of S. Choose a distinguished triangle (X, Y, Z, f, g, h). Next, choose a distinguished triangle $(X', Y', Z, f', g', t[1] \circ h)$ (here we use TR1 and TR2). By MS5, MS6 (and TR2 to rotate) we can find the dotted arrow in the commutative diagram

$$\begin{array}{ccccccc} X & \longrightarrow & Y & \longrightarrow & Z & \longrightarrow & X[1] \\ \downarrow{\scriptstyle t} & & \vdots{\scriptstyle s'} & & \downarrow{\scriptstyle 1} & & \downarrow{\scriptstyle t[1]} \\ X' & \longrightarrow & Y' & \longrightarrow & Z & \longrightarrow & X'[1] \end{array}$$

with moreover $s' \in S$. This proves LMS2. The proof of RMS2 is dual. $\square$